two DNA molecules are formed and each of these two molecules will have one parental strand and one newly synthesized strand.

6.2.4 The dispersive model

This model states that the parental DNA is randomly segmented into multiple double stranded DNA fragments. These fragments undergo replication similar to conservative model. The fragments later reassemble into complete DNA molecules. This model of replication will result in synthesis of two DNA molecules with interspersed parental and newly synthesized double helix.

6.2.5 The Meselson-Stahl experiment and the conformation of semiconservative model

The semiconservative model of DNA replication was confirmed by Matthew Meselson and Franklin Stahl in 1958. They took advantage of the fact that though ^{14}N isotope of nitrogen is normally present in DNA, the presence of the heavier ^{15}N isotope doesn't affect the function of DNA.

In this experiment, three different batches of the bacterium *E. coli* were grown. The first batch was grown with the medium containing ^{15}N as the only source of nitrogen for several generations. The bacteria took up this ^{15}N for growth and eventually all the nitrogen in the DNA were ^{15}N isotope. The *E. coli* were then switched suddenly to the medium containing the lighter ^{14}N as the only source of nitrogen, so as the bacterium divides these lighter nitrogen gets incorporated into the DNA. After switching to the light ^{14}N medium, sample DNA were isolated for every generation of *E. coli*, starting from generation zero (ie., every 20 mins). The DNA extracted subjected to caesium chloride (CsCl) density gradient centrifugation. This method of centrifugation separates the DNA into bands according to their density, with heavier DNA band at the bottom and progressively lighter DNA bands at the top.

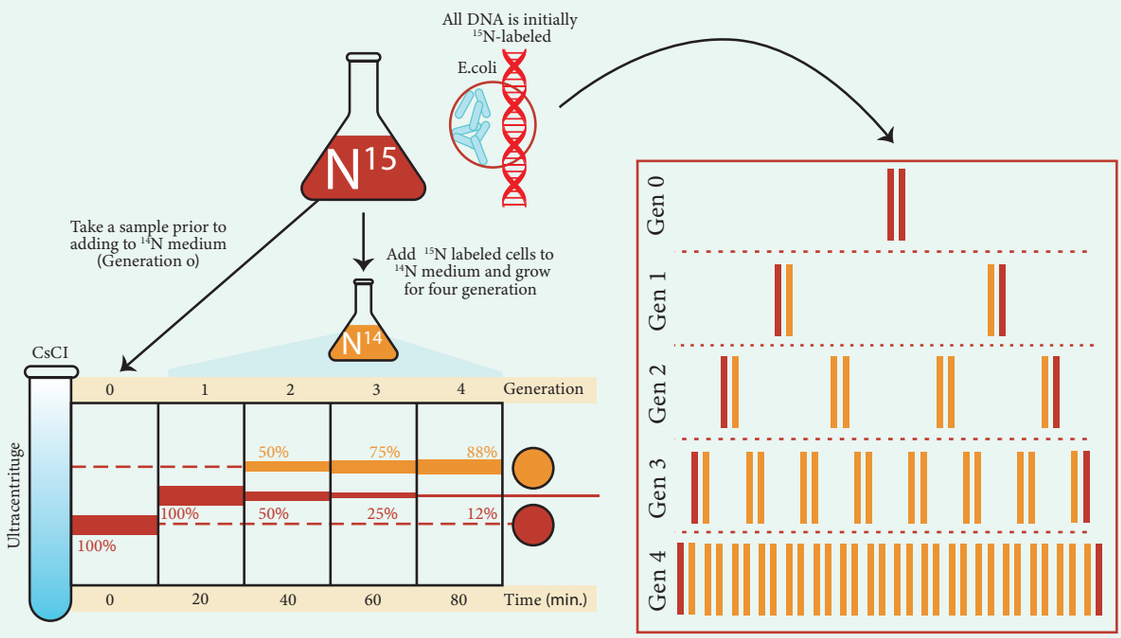


Figure 3 The schematic representation of Meselson-Stahl experiment

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